

LECTURE-24

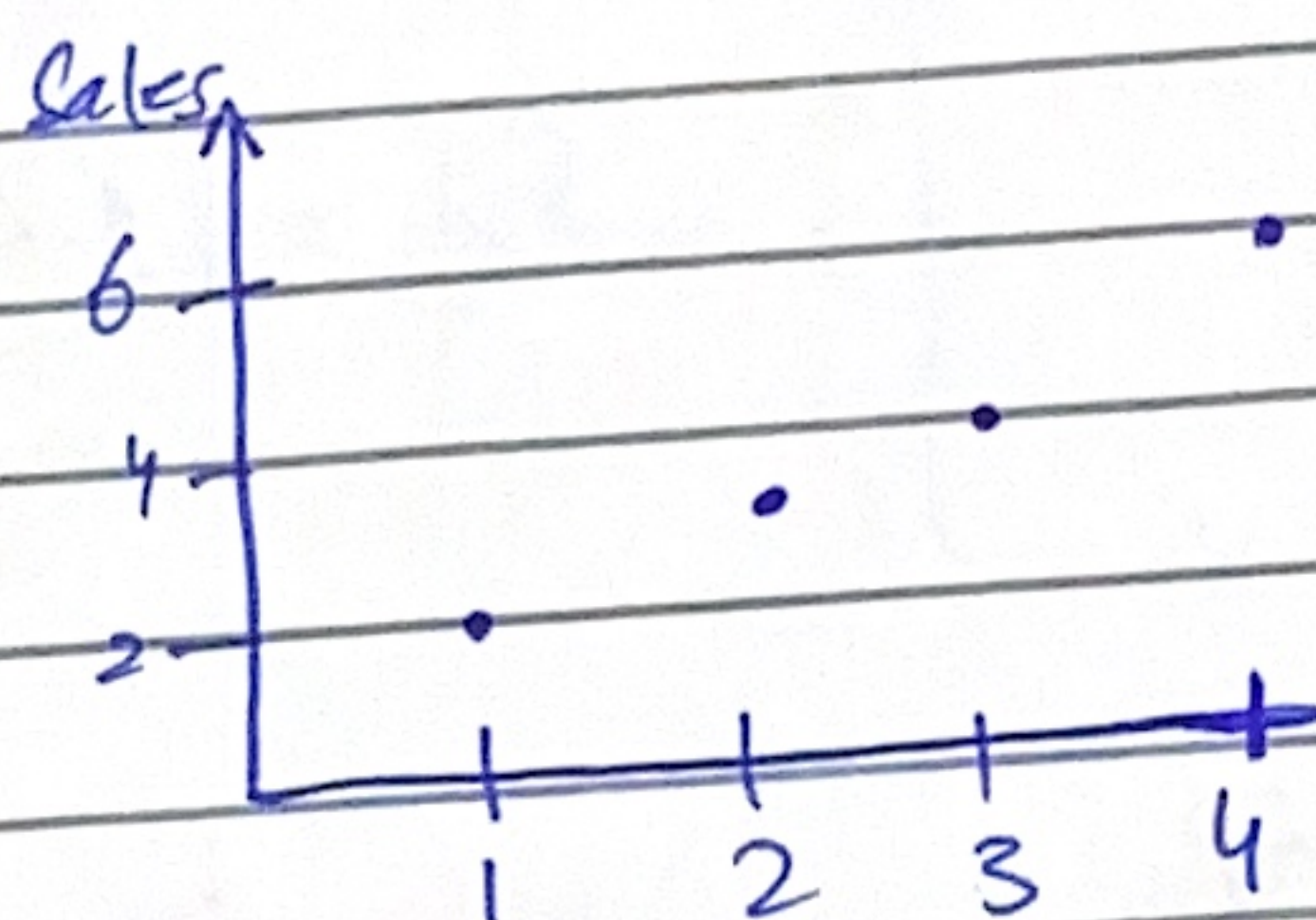
wednesday
6/5/26.

Correlation Analysis:

Q: A data analysts want to examine the relationship
b/w Sales & Adv. Expenditure.

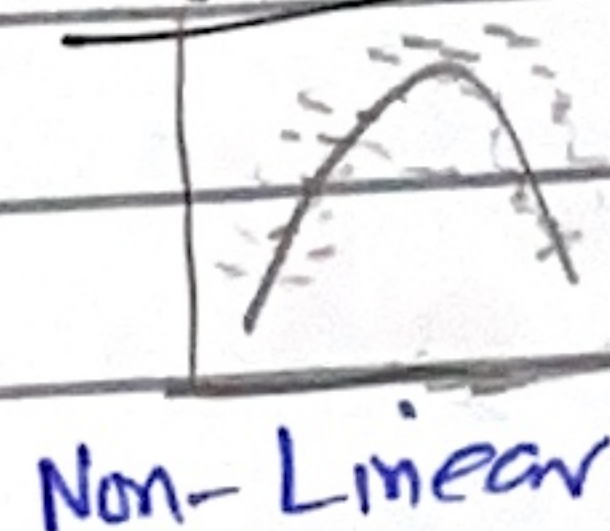
Sales	Adv. Expenditure
2	1
3	2
4	3
6	4

(i) Scatter Diagram:



Positive Linear
Relationship b/w
Sales & Adv. Exp.

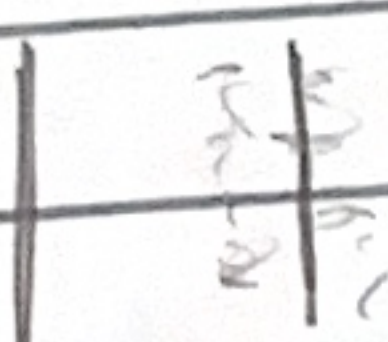
Types:



Non-Linear



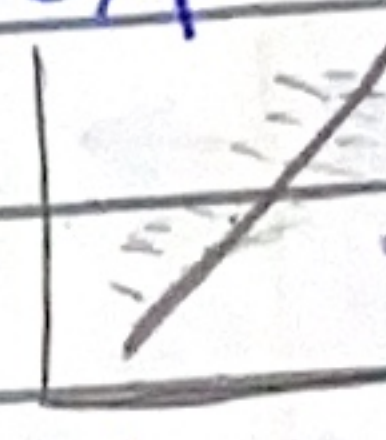
No-Relation



No-Relation



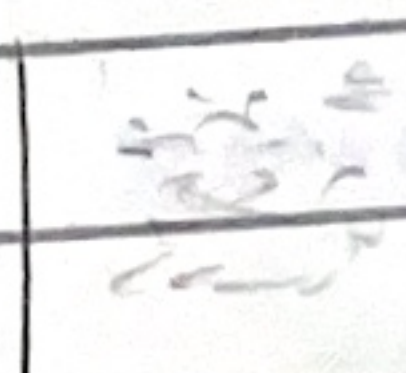
Non-Linear



+ Linear



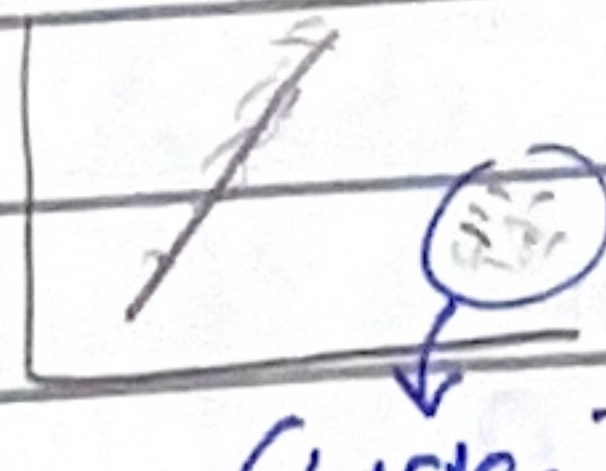
-ve Linear



No Relation



Outlier



Clustering

Covariance is only found for strength of Linear relations

□ To find Strength of Relation b/w two variables.

⇒ In Population:

$$\text{Cov}(X, Y) = \sigma_{xy}$$

$$\sigma_{xy} = \frac{1}{N} \left[\sum XY - \frac{(\sum X)(\sum Y)}{N} \right]$$

⇒ In Sample: $\text{Cov}(X, Y) = S_{xy}$

$$S_{xy} = \frac{1}{n} \left[\sum XY - \frac{(\sum X)(\sum Y)}{n} \right]$$

Difference is 'N' and 'n'

(ii) Find Covariance b/w Sales & Adv. Exp.

X	Y	XY
Sales	Adv. Exp	
1	2	2
2	3	6
3	4	12
4	6	24
$\sum X = 10$	$\sum Y = 15$	$\sum = 44$

$$\text{Cov}(X, Y) = S_{xy} = \frac{1}{4} \left[44 - \frac{(10)(15)}{4} \right]$$

$$S_{xy} = 1.625$$

NOTE	S_{xy}	Relationship
closer to 0	0	⇒ No relationship
	+ve	⇒ +ve relationship
	-ve	⇒ -ve "

only tells direction of B.S.

For we'll use correlation.

⇒ Correlation: Tells the strength of relationship as Covariance only showed the direction of B.S.

• Population: $\text{Corr}(X, Y) = \rho_{xy}$

• Sample: $\text{Corr}(X, Y) = r_{xy}$

$$r_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X) \cdot \text{var}(Y)}}$$

$$= \frac{\overset{\text{Covariance}}{S_{xy}}}{\sqrt{S_x^2 \cdot S_y^2}}$$

standard deviation

(iii) Find correlation coefficient: ^{old formula.}

X^2	Y^2
1	4
4	9
9	16
16	36
$\Sigma = 30$	$\Sigma = 65$

$$S.D.: S_x = \frac{1}{n} \left[\Sigma x^2 - \frac{(\Sigma x)^2}{n} \right]$$

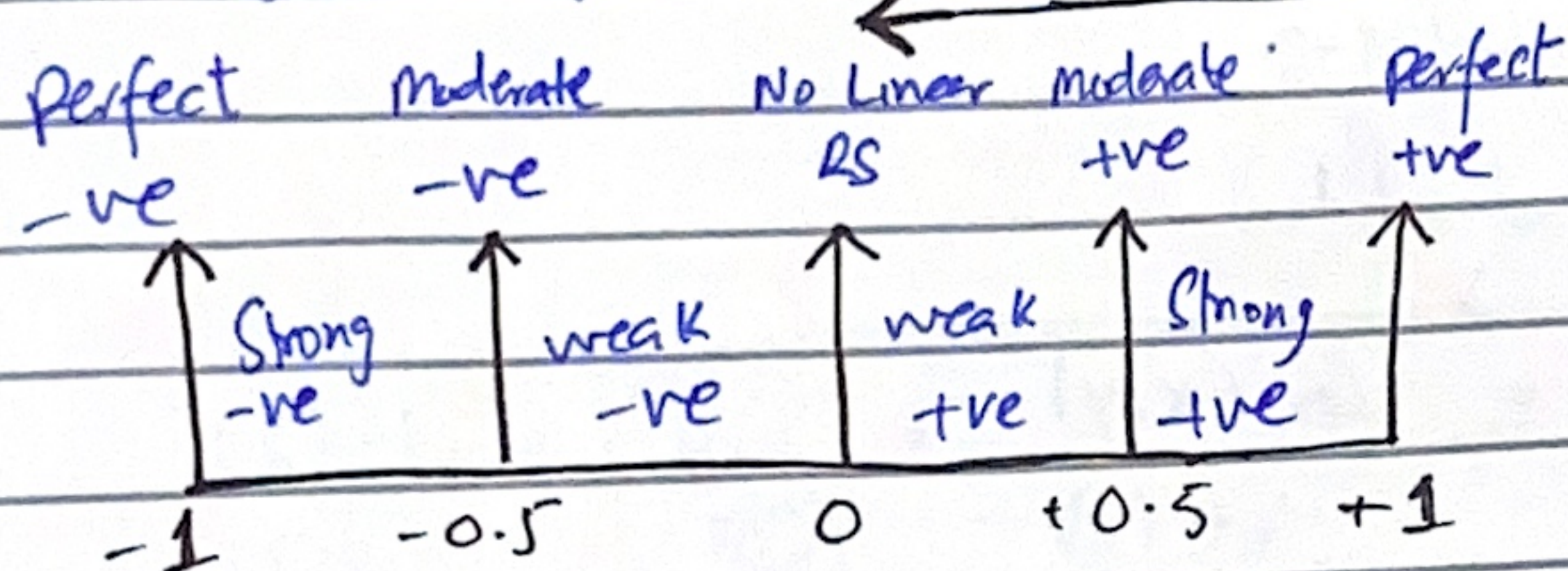
$$= \frac{1}{4} \left[30 - \frac{(10)^2}{4} \right]$$

$$S_x^2 = 1.25$$

$$S_y = \frac{1}{4} \left[65 - \frac{(15)^2}{4} \right]$$

$$S_y^2 = 2.1875$$

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_x^2 \cdot S_y^2}} = \frac{1.625}{\sqrt{1.25 \times 2.1875}} = 0.98$$



∴ we've a Strong ~~the~~ relationship b/w sale & expenditure.

⇒ Regression: We need to now predict a future value on basis of current info.

• Here X & Y cannot be any variables like in "correlation" or "covariance". X is independent & Y is dependent.
 Explanatory, Predictor, Regressor ← Explained, Predictand, Regressand

• Deterministic model: Single output from single input having same fixed values. eg: Area = πr^2 . (Here r is affecting it).
 $r=1$ pr Area = π ; as area has just 'r' affecting it.

• Probabilistic model: Output depends upon more parameters like weight = $20 + 30(\text{age})$. Now har kesi ka weight same rahi hoga jiski age same hogi ∴ more factors are needed to compute weight.

Deterministic: $Sales = \beta_0 + \beta_1 \cdot (Expenditure)$

Probabilities: $Sales = \beta_0 + \beta_1 \cdot (Expenditure) + \epsilon$
regression uses this
dist Bin: all other factors effecting sales apart from "Expenditure"

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

y is dependent upon 3 variables that we are concerned with

• To find values of $\beta_0, \beta_1, \beta_2, \beta_3$; we use "Method of Least Square".

Q: Same ques. as page 1 of Lecture.

(a) Estimate regression eq. sales & Exp!

→ Linear Simple reg. model → $y = \beta_0 + \beta_1 x_1 + \epsilon$

→ Estimated Linear Simple reg. model →

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

~~$\beta_0 = 3.15$~~ mean of y

$$\bar{y} = 15/4 = 3.75$$

$$\bar{x} = 10/4 = 2.5$$

$$\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{S_{xy}}{(S_x)^2} = \frac{1.625}{1.25} = 1.3$$

old calculations

$$\hat{y} = 0.5 + (1.3)x \quad \leftarrow \text{answer}$$

Standard Error of Estimation: To see how good is our regression eq.
Population: $\sigma_{y \cdot x}$ → means y is depend't on x.

$$\text{Sample: } S_{y \cdot x} = \sqrt{\frac{\sum Y^2 - b_0 \sum Y - b_1 \sum XY}{n-2}}$$

• Lower the answer; better the fit of model. Higher the Standard error of estimation; worse the fit of model

- Standard error's answer just gives direction kay acha model hai yaan bura but didn't tell strength just like co-variance

- Therefore for strength of standard error of estimation; we use Coefficient of Determination (just like co-relation)
↓ correlation of sample
 $R^2 = (r_{xy})^2$

$$0 \leq R^2 \leq 1$$

- $0 \rightarrow 0.5$: Poor fit
 $0.5 \rightarrow 0.70$: Medium fit
 $0.70 \rightarrow 1$: good fit.